

Dr. Sapna Mishra

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RESEARCH INTERESTS

- Satellite–host galaxy interactions.
- Magellanic Clouds and Stream.
- Galaxy evolution in the Local Group.
- Circumgalactic and Intergalactic Medium.
- Outskirts of the Galaxy Cluster.
- AGN feedback.
- Quasar absorption line spectroscopy.

ACADEMIC POSITIONS

- **Postdoctoral Fellow, 2023–present**, Milky Way Halo Research Group, Space Telescope Science Institute (STScI), Baltimore, USA.
- **Postdoctoral Fellow, 2021–2023**, Inter-University Centre for Astronomy & Astrophysics (IUCAA), Pune, India.
- **Post-Thesis Submission Fellow, 2020–2021**, Aryabhata Research Institute of observational sciences (ARIES), Nainital, India.

EDUCATION

- **Ph.D., Astronomy, 2015–2020**, ARIES, Nainital, India. *Thesis: Probing the Environment of AGN Based on Their Feedback Processes*; Advisor: Prof. Hum Chand.
- **Pre-Ph.D. Coursework in Astronomy, 2014–2015**, ARIES, Nainital, India.
- **Master of Science (MSc), Physics, 2012–2014**, Department of Physics & Astrophysics, Delhi University (DU), India.
- **Bachelor of Science (BSc), Physics Honors, 2009–2012**, Miranda House College, DU, India.

RESEARCH GRANTS

- **2026:** JWST Cycle 5 (PI: T. Fischer), [GO–10815](#).
- **2025:** HST Cycle 33 (PI: S. Mishra), [GO+AR–18076](#), *Funding* ~ \$99K.
- **2025:** HST Cycle 33 (PI: S. Mishra), [GO+AR–18072](#), *Funding* ~ \$84K.
- **2025:** HST Cycle 33 (PI: A. Fox), [AR–18149](#), *Funding* ~ \$29K.
- **2024:** HST Cycle 32 (PI: S. Mishra), [GO–17757](#), *Funding* ~ \$110K.
- **2024:** Simons Foundation — Support to attend the CGM Workshop, Aspen Center for Physics; *Funding* ~ \$3,000.
- **2023:** [FONDECYT Postdoctoral Fellowship 2023](#) (Chile), *Program* #3230509, *Funding* \$110,400.
- **2022:** [Milano–Bicocca Research Grant \(Type A2\)](#) (Italy).

TELESCOPE TIME AS PRINCIPAL INVESTIGATOR

- **VLT/UVES (Cycle P117): 35 hours**, *The Distance to the Magellanic Stream Using RR Lyrae Stars*, *PID: 117.2A4J*.
- **HST/COS (Cycle 33): 30 orbits**, *Unveiling the Circumgalactic Medium of M33*, *PID: GO+AR-18076*.
- **HST/COS (Cycle 33): 22 orbits**, *The Fate of the Leading Arm of the Magellanic Stream*, *PID: GO+AR-18072*.
- **HST/COS (Cycle 32): 29 orbits**, *Probing the Front-Side of the Circumgalactic Medium of the Large Magellanic Cloud*, *PID: GO-17757*.
- **VLT/FORS2 (Cycle P109): 17 hours**, *MgII Tomography of Cluster Outskirts Using 11 Background Quasars*, *PID: 109.23G6*.
- **Devasthal Optical Telescope (DOT, 3.6m, Cycle 22A): 2 Nights**, *NIR Spectroscopy of Post-starburst Galaxies to Probe Obscured Star Formation and Stellar Populations*, *PID: DOT-2022-C1-P18*.
- **DOT (Cycle 21A): 4 Nights**, *Probing the Connection Between Emission and Absorption Outflows in IR-bright BAL Quasars*, *PID: DOT-2021-C1-P32*.

- **DOT (Cycle 2018A): 16 Hours**, *Resolving the Narrow Emission-Line Region of the Quadruply Imaged Quasar RXS J113155.4–123155*, PID: P325-2018A.
- **DOT (Cycle 2017A): 4 Nights**, *Infrared Properties of Jet-dominated BALQSOs*, PID: P31-2017A.
- **Himalayan Chandra Telescope (HCT, 2m, Cycle 21-C2): 3 Nights**, *Probing Spectral Variability of X-ray Bright High-ionization BAL Quasars*, PID: HCT-2021-C2-P56.
- **HCT (Cycle 21-C1): 3.5 Nights**, *Intranight Monitoring of Blazar Counterparts of BAL Quasars*, PID: HCT-2021-C1-P52.
- **HCT (Multiple cycles): ≈ 12 Nights**, *Probing Environments of Emerging Broad Absorption Line Quasars*, PIDs: HCT-2020-C2-P27; HCT-2020-C1-P170; HCT-2019-C3-P117.

AS CO-INVESTIGATOR

- **JWST (Cycle 5): 14.6 hours**, *Following the Shocks: JWST IFU Spectroscopy of NGC 4258's Radio Arcs*; PID: GO-10815.
 - **NRAO/GBT (Cycle 26A): 370 hours**, *Mapping Cool Clouds in the Milky Way's Nuclear Wind*; PID: GBT26A-224.
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PRESS COVERAGE

- [Mishra et al. \(2024b\)](#) featured as a question on the **American TV quiz show *Jeopardy!*** (Feb 12, 2025).
 - [NASA's Hubble Sees Aftermath of Galaxy's Scrape with Milky Way](#) — NASA press release.
 - [Hubble sees aftermath: Encounter blew away most of smaller galaxy's gaseous halo](#) — ESA/Hubble press release.
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SKILLS

- **Observing Experience:** ~ 100 nights with 1–4m ground-based telescopes.
 - **Techniques:** Spectroscopy (expert, 8 first-author papers), Photometry (proficient, 2 first-author papers), Astrometry (calibration-level experience, 1 proceeding).
 - **Spectrographs Used:** HST/COS; VLT (UVES, X-shooter, FORS2); Keck/LRIS; SDSS/BOSS; SAO/SCORPIO; HST/HFOSC, DOT/DFOT.
 - **Programming Languages:** Python, IDL, C, C++, ecl-IRAF scripts, Unix shell scripts.
 - **Web Programming:** PHP, MySQL, HTML.
 - **Astronomy Software:** IRAF, DAOPHOT, CLOUDY, TOPCAT, ESOReflex/EsoRex, Gasgano, vpfitt.
 - **Large-Dataset Queries:** SQL CasJobs queries for SDSS, HST-MAST, SIMBAD, NED.
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PROFESSIONAL & ACADEMIC SERVICE

- **Panel Support Scientist**, HST/TAC, Cycle 34, STScI (2026).
 - **LOC**, Spring Symposium, STScI (2026).
 - **Panelist**, National Science Foundation (NSF), Remote (2026).
 - **Reviewer & Mentor**, Space Astronomy Summer Program (SASP), STScI (2026).
 - **Member**, American Astronomical Society.
 - **Referee**, ApJ, ApJL.
 - **Member**, Working Group on the High-Resolution Multi-Object Spectrograph (HRMOS) for the VLT.
 - **Reviewer**, Chambliss Student Poster Competition, 247th AAS Meeting (2026).
 - **Reviewer**, Space Astronomy Summer Program (SASP), STScI (2025).
 - **Panel Support Scientist**, HST/TAC, Cycle 33, STScI (2025).
 - **Organizing Committee Member**, STScI HotSci Colloquium Series (2024).
 - **Panel Support Scientist**, HST/TAC, Cycle 32, STScI (2024).
 - **Panel Support Scientist**, JWST Telescope Allocation Committee (TAC), Cycle 3, STScI (2024).
 - **Service Observer**, Devasthal Optical Telescope (3.6m) — conducted observations on behalf of external proposers during the COVID-19 period.
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MENTORING

- **June 2026–present: Noela Weramundi**, Space Astronomy Summer Program (SASP-2026).
 - **Feb 2025–present: Zhibin You**, Undergraduate student, Johns Hopkins University.
Research project: “*Probing the Small-Scale Routly–Spitzer Effect in the Milky Way ISM.*”
 - **Aug 2025–present: Khushi Mehta** (remote), Central University of Himachal Pradesh (India).
Master’s dissertation: “*Probing the Circumgalactic Medium of Virgo Cluster Galaxies.*”
 - **2023–2025: Ritish Kumar** (remote), Central University of Himachal Pradesh (India).
Graduate research project: “*On the Incidence of Weak and Strong MgII Absorbers Toward Flat and Steep Spectrum Radio Quasars.*” Published in MNRAS (2025).
 - **2017–2018: Sri Devi and Parth Nair**, Master’s students in Physics.
Summer dissertation projects: “*Photometric and Spectroscopic Variability of BAL Quasars.*”
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TEACHING & OUTREACH

- **Graduate-Level Instructor, IUCAA, India** — Taught three online graduate-course lectures on Astronomical Instruments (June 2022).
- **Lecturer, ARIES Training School in Observational Astronomy (ATSOA)** — Delivered spectroscopy lectures (theory and observation) in 2016, 2017, 2018, and 2019.
- **Hands-on Data-Reduction Instructor, ATSOA** — Provided 14-day intensive training in optical data reduction (2016–2019).
- **Instructor, TMT Workshop on Large Telescope Data Handling (IUCAA)** — Conducted training in high-resolution UVES spectral data reduction (Jan 15–27, 2017).
- **Mentor for Incoming Ph.D. Students, ARIES, India** — Guided students in telescope operations, observing techniques, and data handling.

Outreach & Public Engagement

- Astronomy for Non-Astronomers, STScI, June 21, 2026 (invited public talk).
 - Delivered invited **public talks** at “Astronomy on Tap” (Baltimore, 2025) and undergraduate outreach lectures.
 - Led **public sky-gazing sessions** using 1–2m class telescopes at ARIES and Devasthal observatories, India.
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AWARDS & HONORS

- **2023: FONDECYT-2023 Postdoctoral Prize Fellowship**, Chile.
 - **2022: Milano-Bicocca Research Grant (Type A2)**, Italy.
 - **2014: All-India “Graduate Aptitude Test in Engineering” (GATE)**, India.
 - **2012: All-India “Joint Admission Test for Masters” (JAM)**, India.
 - **2012: Selected in the top 10% of graduate-level students**, Delhi University.
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SCIENTIFIC PRESENTATIONS

- **2026:**
 - FOGGIE Retreat Meeting, Mt. Washington, Baltimore, June 1, 2026.
 - Astronomy for Non-Astronomers, STScI, June 21, 2026 (invited public talk).
 - Galaxies and AGN Seminar and Journal Club, JHU & STScI, Feb 24, 2026 (invited, declined).
 - Centre for Particle Physics and Astronomy, Royal Holloway, University of London, Jan 28, 2026 (invited, declined).
 - 247th American Astronomical Society Meeting, Phoenix, Arizona, Jan 2026 (talk).
 - 247th American Astronomical Society Meeting, Phoenix, Arizona, Jan 2026 (splinter session).
- **2025:**
 - Aix-Marseille Université, “Workshop on the Network for Ultraviolet Astronomy,” Oct 2025.
 - Space Telescope Science Institute, “HotSci Colloquium Series,” July 2025.
 - Montana State University, “XMCII: Milky Clouds over Yellowstone,” May 2025.
 - Space Telescope Science Institute, “Spring Symposium,” May 2025 (poster/flash talk).
 - Summer Conference 2025, Ninth Edition, May 2025 (remote, invited talk).

• **2024:**

- Aspen Center for Physics, “Holistic Picture of the CGM,” September 2024.
- Center for Astrophysics | Harvard & Smithsonian, “Multiphase Madness,” August 2024.
- Space Telescope Science Institute, “Spring Symposium,” April 2024.
- Flatiron Institute, “XMCI: Milky Clouds over Manhattan,” February 2024.
- Space Telescope Science Institute, “Galaxy/AGN Journal Club,” January 2024.
- Space Telescope Science Institute, “CoolSci,” January 2024.

- **2023:** IUCAA, India, “Galactic Inflows and Outflows on All Scales,” February 2023.

• **2022:**

- Università Milano-Bicocca, Milan, “What Matter(s) Around Galaxies,” September 2022.
- IUCAA, India, “Monthly Last Friday Talk Series,” January 2022.

• **Pre-2020:**

- IISER Tirupati, India, “Astronomical Society of India Meeting,” March 2020 (poster).
- Department of Physics & Astrophysics, Delhi University, “Departmental Talk,” October 2019 (invited).
- IUCAA, India, “Recent Trends in the Study of Compact Objects: Theory and Observations (RETICO-IV),” April 2019 (poster).
- ARIES, India, “ARIES Training School in Observational Astronomy (ATSOA),” March 2019.
- Institut d’Astrophysique de Paris (IAP), Paris, France, “Massive Black Holes in Evolving Galaxies: From Quasars to Quiescence,” May 2018 (poster/flash talk).
- ARIES, India, “ARIES Training School in Observational Astronomy (ATSOA),” March 2018.
- Département d’Astrophysique, Géophysique et Océanographie, Université de Liège, Belgium, December 2017 (invited).
- IUCAA, India, “Thirty Meter Telescope (TMT) Conference,” January 2017.
- ARIES, India, “ARIES Training School in Observational Astronomy (ATSOA),” March 2017.
- ARIES, India, “Belgo-Indian Network for Astronomy and Astrophysics (BINA),” November 2016 (poster/flash talk).
- ARIES, India, “ARIES Training School in Observational Astronomy (ATSOA),” February 2016.
- ARIES, India, “Tuesday Seminar Series,” May 2016; February 2017.
- IUCAA, India, “Cloudy Workshop,” September 2015.

WORKSHOPS AND SCHOOLS

- AstroSat Data Analysis Workshop, August 8–11, 2017, ARIES, Nainital, India.
 - TMT Workshop on Large Telescope Data Handling, Jan 15–27, 2017, IUCAA, Pune, India.
 - Extragalactic Relativistic Jets: Cause and Effect; FERMI Satellite Data Reduction School, ICTS Bangalore, October 14–21, 2015.
 - Cloudy Workshop, September 21–26, 2015, IUCAA, Pune, India.
 - Workshop on Radio Data Reduction, Radio Astronomy School 2015 (RAS), August 31, 2015, NCRA, Pune, India.
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LIST OF PUBLICATIONS

First-Author Publications

1. **Mishra, Sapna**; Khaire, Vikram; Pallikara, Romeo; Narayanan, Anand; Fox, Andrew (minor revision, ApJ) [“Discovery of Weak O VI Absorption in Underdense Regions of the Low-Redshift Intergalactic Medium.”](#)
2. **Mishra, Sapna**; Fox, Andrew; Smoker, J.; Lucchini, Scott; D’Onghia, Elena; 2025, ApJ, 984, 104. [“The Distance to the Magellanic Stream: Constraints from Optical Absorption along Stellar Sightlines.”](#)
3. **Mishra, Sapna**; Fox, Andrew; Krishnarao, Dhanesh; Lucchini, Scott; D’Onghia, Elena; Cashman, Frances; Barger, Kathleen; Lehner, Nicolas; Tumlinson, Jason; 2024, ApJL, 976, L28. [“The Truncated Circumgalactic Medium of the Large Magellanic Cloud.”](#)
4. **Mishra, Sapna**; Muzahid, Sowgat; Dutta, Sayak; Srianand, Raghunathan; Charlton, Jane; 2024, MNRAS, 527, 3858. [“Characterizing Cool, Neutral Gas and Ionized Metals in the Outskirts of Low- \$z\$ Galaxy Clusters.”](#)

5. **Mishra, Sapna**; Muzahid, Sowgat; 2022, ApJ, 933, 229. “*Discovery of a Cool, Metal-rich Gas Reservoir in the Outskirts of $z \approx 0.5$ Clusters.*”
6. **Mishra, Sapna**; Gopal-Krishna; Chand, H.; Chand, K.; Kumar, A.; Negi, V.; 2021, MNRASL, 507, L46. “*A Search for Blazar Activity in Broad-Absorption-Line Quasars.*”
7. **Mishra, Sapna**; Vivek, M.; Chand, H.; Joshi, R.; 2021, MNRAS, 504, 3187. “*Appearance versus Disappearance of Broad Absorption Line Troughs in Quasars.*”
8. **Mishra, Sapna**; Krishna, G.; Chand, H.; Chand, K.; Ojha, V.; 2019, MNRASL, 489, L42. “*Are There Broad Absorption Line Blazars?*”
9. **Mishra, Sapna**; Chand, H.; Krishna, G.; Joshi, R.; Shchekinov, Y. A.; Fatkhullin, T. A.; 2018, MNRAS, 473, 5154. “*On the Incidence of MgII Absorbers along Blazar Sightlines.*”

Second-Author Publications

1. † Zhibin, You; **Mishra, Sapna**; Fox, Andrew (to be submitted to ApJ). “*Probing the Small-Scale Routly–Spitzer Effect in the Milky Way Halo.*”
2. ‡ Fox, Andrew; **Mishra, Sapna**; Cashman, Frances; French, David; Richter, Philipp; Bordoloi, Rongmon; Lehner, Nicolas; Tumlinson, Jason; Borthakur, S.; 2026, ApJ, 999, 107. “*Low-Metallicity Gas on the Outskirts of the Local Group: The Circumgalactic Medium of Sextans B.*”
3. † Kumar, Ritish; **Mishra, Sapna**; Chand, Hum; 2025, MNRAS, 542, 119. “*On the Incidence of Weak and Strong MgII Absorbers Toward Flat and Steep Spectrum Radio Quasars.*”

Co-Author Publications

1. Lucchini, Scott; Han, Jess; **Mishra, Sapna**; Fox, Andrew; 2026, ApJ, 1002, 14. “*The LMC Corona Favors a First Passage*”
2. ‡ Dutta, Sayak; Muzahid, Sowgat; Schaye, Joop; **Mishra, Sapna**; Chen, Hsiao-Wen; Johnson, Sean; Wisotzki, Lutz; Cantalupo, Sebastiano; 2024, MNRAS, 528, 3745. “*MUSEQuBES: Mapping the Distribution of Neutral Hydrogen around Low-Redshift Galaxies.*”
3. Gopal-Krishna; Chand, K.; Chand, H.; Negi, V.; **Mishra, Sapna**; Britzen, S.; Bisht, S.; 2023, MNRASL, 518, L13. “*Intranight Optical Variability of Low-mass Active Galactic Nuclei: A Pointer to Blazar-like Activity.*”
4. ‡ Kumar, B.; Negi, V.; Ailawadhi, B.; **Mishra, Sapna**; Pradhan, B.; Misra, K.; Hickson, P.; Surdej, J.; 2022, JApA, 43, 10. “*Upcoming 4m ILMT Facility and Data Reduction Pipeline Testing.*”
5. ‡ Chand, K.; Gopal-Krishna; Omar, A.; Chand, H.; **Mishra, Sapna**; Bisht, S.; Britzen, S.; 2022, MNRASL, 511, L13. “*Intranight Variability of Ultraviolet Emission from Powerful Blazars.*”
6. ‡ Ojha, V.; Chand, H.; Gopal-Krishna; **Mishra, Sapna**; Chand, K.; 2020, MNRAS, 493, 3642. “*Comparative Intranight Optical Variability of X-ray and γ -ray Detected Narrow-Line Seyfert 1 Galaxies.*”

†: Primary mentor; ‡: Significant contribution in analysis/observations.

CONFERENCE PROCEEDINGS (refereed)

4. Fox, Andrew; Kriss, Jerry; Richter, Philipp; Shull, J. Michael; Cashman, Frances; **Mishra, Sapna**; et al. 2026, White Paper responding to the STScI request for community input. “*High-S/N Quasar Observations with HST/COS: Deep Fields for Spectroscopy*”
3. Vivek, M.; Nair, Akhil; **Mishra, Sapna**; Proceedings of the IAU Symposium No. 378, 2024. “*AGN Outflows and Their Variability.*”

2. **Sapna Mishra**; Chand, H.; et al. 2018, Bulletin de la Société Royale des Sciences de Liège, 87, 325. *“Revisiting the Incidence of MgII Absorbers along Blazar Sightlines.”*
 1. Chand, Hum; Rakshit, Suvendu; Jalan, Priyanka; Ojha, Vineet; Srianand, Raghunathan; Vivek, Mariappan; **Sapna Mishra**; et al. 2018, Bulletin de la Société Royale des Sciences de Liège, 87, 291. *“Probing the Central Engine and Environment of AGN Using ARIES 1.3-m and 3.6-m Telescopes.”*
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